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SUPERSEDING
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23 December 1965

MILITARY SPECIFICATION

N-PHENYLMORPHOLINE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers one grade of N-phenylomorpholine.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation - Packaging, Methods of
MIL-P-14232	Parts, Equipment, and Tools for Army Materiel, Packaging and Packing of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publication required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing Materials.

ASTM D 1303 — Total Chlorine in Vinyl Chloride Polymers and Copolymers

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample sahll meet all the requirements of this specification.

3.2 Chemical analysis. The chemical analysis of the material shall be as specified in Table I.

Table I. Chemical analysis

Characteristics	Minimum	Maximum
N-phenylmorpholine, total alkalinity, percent	99.0	--
N-phenylmorpholine, tertiary amine, percent	98.0	--
Chlorine, percent	-	0.10
Melting point, degrees Celsius (C)	50	54
Moisture, percent	-	0.10

3.3 Workmanship. The N-phenylomorpholine shall be uniform in quality, free from foreign materials, and manufactured under controlled conditions and procedures standard in the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification

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where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptance inspection. Acceptance inspection shall, at the option of the Government, be performed either by the supplier or by a Government representative. In either case, responsibility for final acceptance rests with the Government representative.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of N-phenylmorpholine. The preproduction sample shall be subjected to all the examinations and tests specified herein. Unless otherwise specified, the Government will perform the examinations and tests for preproduction sample acceptance at the contractor's plant. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Lot. A lot shall consist of material produced at one plant with no change in formulation or process. If manufacture is by batch process, each batch shall constitute a lot. A batch shall be as defined in 6.3.

4.4 Acceptance sampling. The number of containers to be chosen at random for acceptance sampling shall be equal to the square root of the total number of containers in the lot. If the number thus obtained is not a whole number, the number of containers to be sampled shall be increased to the next higher whole number. In no case, however, shall the number of containers to be sampled be less than seven (unless there are less than seven containers in the lot; in which case, each container shall be sampled).

4.4.1 Primary sample. From each selected container, a sample shall be taken from three or more places throughout the container. The total weight of the samples taken from each container shall weigh at least 50 grams (g). Each sample thus taken shall be mixed thoroughly, placed in a clean, dry container, and labeled to identify the material name, original container designation, contract number, and lot number.

4.4.2 Composite sample. Each primary sample shall be subdivided to prepare a composite sample (not in excess of 500 g). Primary material not used shall be returned to the primary sample container. After mixing the composite sample thoroughly, the composite sample shall be placed in a clean, dry container and sealed. The composite sample shall be identified with the material name, container designations, contract number, and lot number. All specified chemical and physical tests shall be made on this composite sample representing the lot. Failure of the composite sample to pass all of the tests herein shall result in rejection of the lot represented.

4.5 Classification of tests. Inspection and testing of N-phenylmorpholine shall be classified as follows

a. Preproduction tests.

b. Quality conformance tests.

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4.6 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all examinations and tests specified herein.

4.7 Quality conformance tests. Quality conformance tests for acceptance of the N-phenylmorpholine shall consist of the following tests:

Characteristics	Tests
N-phenylmorpholine, total alkalinity	4.8.1
N-phenylmorpholine, tertiary amine	4.8.2
Chlorine	4.8.3
Melting point	4.8.4
Moisture	4.8.5

4.8 Tests. The following procedures shall be used to determine that the requirements of this specification have been met. Any changes in the test procedures employed shall be subject to prior approval of the procuring activity. Unless otherwise specified, all tests shall be run in duplicate. The average of the two results shall be taken as the test result.

4.8.1 Percent N-phenylmorpholine by total alkalinity.

4.8.1.1 Procedures.

WARNING

This procedure involves the use of perchloric acid. Personnel using this reagent must be made aware of its potential hazards.

a. Transfer approximately 5 g of the molten sample to a tared 100-milliliter (ml), glass-stoppered volumetric flask. Reweigh the flask to the nearest 0.0001 g. The difference in weight is the weight of the sample.

b. Add 50 ml of redistilled acetonitrile to the flask and swirl to effect complete solution. Dilute to the mark with acetonitrile and mix thoroughly.

c. Into each of two 500-ml, glass-stoppered Erlenmeyer flasks, pipet 10 ml of the sample dilution.

d. Pipet 10 ml of redistilled acetonitrile into each of two additional 500-ml Erlenmeyer flasks and reserve for the blank determination.

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e. To each of the flasks add 100 ml of glacial acetic acid which has been neutralized to the first green color using 0.1N perchloric acid ($HClO_4$) and 1.0 percent crystal violet indicator in acetic acid.

f. Add one or two drops of crystal violet indicator to each flask and titrate with standard 0.1N $HClO_4$ to the first green endpoint.

$$\text{Percent N-phenylmorpholine equals } \frac{(A \text{ minus } B)\ N \text{ times } 16.32}{C \text{ times } 0.1}$$

Where: A equals volume of N normal $HClO_4$ required for sample, ml.

B equals volume of N normal $HClO_4$ required for blank, ml.

C equals weight of sample taken, g.

N equals normality of $HClO_4$.

4.8.1.2 Acceptance criteria. For the lot represented to pass the N-phenylmorpholine test, the value obtained for the percent N-phenylmorpholine by total alkalinity shall be not less than the value specified in 3.2.

4.8.2 Percent N-phenylmorpholine by tertiary amine.

4.8.2.1 Procedure.

a. Into each of two 500-ml, glass-stoppered Erlenmeyer flasks, pipet 10 ml of the sample dilution (4.8.1.1b).

b. Into each of two additional 500-ml Erlenmeyer flasks, pipet 10 ml of redistilled acetonitrile and reserve for the blank determination.

c. While swirling slowly, add 20 ml of refined acetic anhydride to each of the blank and sample flasks and allow to stand at room temperature for 5 to 10 minutes.

d. Into each of the flasks, introduce 100 ml of glacial acetic acid which has been neutralized with 0.1N $HClO_4$ to the first green endpoint of crystal violet indicator.

e. Swirl the contents of the flasks, add one or two drops of crystal violet indicator and titrate with standard 0.1N $HClO_4$ to the first green endpoint.

$$\text{Percent N-phenylmorpholine equals } \frac{(A \text{ minus } B)\ N \text{ times } 16.32}{C \text{ times } 0.1}$$

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Where: A equals volume of N normal $HClO_4$ required for the sample, ml.

B equals volume of N normal $HClO_4$ required for the blank, ml.

C equals weight of sample taken, g.

N equals normality of $HClO_4$.

4.8.2.2 Acceptance criteria. For the lot represented to pass the N-phenylmorpholine test, the value obtained in the percent N-phenylmorpholine by tertiary amine shall be not less than the value specified in 3.2.

4.8.3 Chlorine. Determine the amount of chlorine by utilizing the test procedure given by ASTM D 1303, except use ferric alum indicator in place of ferric nitrate indicator and electrical ignition of the Parr bomb in place of ignition by use of a Bunsen burner.

4.8.3.1 Acceptance criteria. For the lot represented to pass the chlorine test, the value obtained for the percent chlorine shall be no greater than the value specified in 3.2.

4.8.4 Melting point.

4.8.4.1 Procedure. Place a few crystals of N-phenylmorpholine on an 18-millimeter (mm) diameter microcover glass. Cover the crystals with an identical cover glass and transfer to the cleaned heating stage of a Fischer-Johns melting point apparatus or equivalent (4.8.4.2). Turn on the apparatus, bring the temperature to 45 degrees Celsius (C) (113 degrees Fahrenheit (F)), and hold the apparatus at that temperature for at least 5 minutes. Then adjust the reading of the current-input dial to give a temperature rise rate of 0.8 to 1.0 degree C (1.4 to 1.8 degrees F) per minute as indicated by the attached thermometer. With the aid of the illuminating and magnifying unit, observe the melting point at which the last crystal of N-phenylmorpholine liquifies.

4.8.4.2 Apparatus. Melting Point Apparatus, Fischer-Johns, available from the Fischer Scientific Company, Pittsburgh, Pennsylvania or equivalent.

4.8.4.3 Acceptance criteria. For the lot represented to pass the melting point test, the value obtained for the temperature of the melting point shall be within the range in 3.2.

4.8.5 Moisture.

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4.8.5.1 Apparatus. The apparatus used for determination of moisture content of the sample shall be an Aquameter, Model KF-2 or KF-3, Beckman Instruments, Inc., Fullerton, California, or an approved equivalent. The Acquameter shall be prepared for operation as described in the technical manual furnished by the manufacturer (Beckman Instruments, Inc.). Use of an alternate equivalent item of equipment approved by the procuring activity will necessitate use of the specific technical manual prepared by the manufacturer.

4.8.5.2 Reagents

a. Karl Fischer reagent. Karl Fischer reagent must have a strength such that each milliliter of Karl Fischer reagent corresponds to 0.0014-0.0023 g of water. Dilute 750 ml of commercially available stabilized Karl Fischer reagent (with water equivalent of 0.005-0.007 g/ml) to 2000 ml with absolute methanol (0.1 percent water, maximum). Mix well and allow to stand overnight before use. Determine the water equivalent (A) of this solution as follows:

1. Use sodium tartrate dihydrate ($Na_2C_4H_4O_6$ $2H_2O$) as a primary standard (with a water content of 15.66 percent) for standardizing Karl Fischer reagent. If the water content value is in question, it may be determined by heating some of the salt at 150 degrees C (302 degrees F) for 3 hours. Should the value (as determined) differ from the theoretical value of 15.66, then the experimental value shall be used in the determination of water equivalent (A) of the Karl Fischer reagent; instead of the 15.66 in the formula below, the factor should be 10P, where P is percentage moisture (as determined). Rapidly transfer 0.090-0.110 g (weighed to the nearest 0.0001 g) of reagent-grade $Na_2C_4H_4O_6$ $2H_2O$ to the titration vessel.

2. Titrate to an endpoint in the same manner as with the sample. (See 4.8.5.3.)

3. Repeat the standardization procedure until three successive results agree within five parts per thousand.

4. If the indicated water equivalent (A) of the Karl Fischer reagent is less than 0.0014 g of water per ml of Karl Fischer reagent, it may be due to the presence of too much water in the absolute methanol used.

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In this case, distill the methanol from metallic calcium or calcium hydride. Passing the methanol through a column of Molecular Sieves, Type 4A, may also reduce the water content of the methanol sufficiently. (Molecular Sieves are a product of the Linde Company, a division of Union Carbide Corporation, New York City, New York.)

Water equivalent (A) equals $\frac{0.1566}{V}$

Where: A equals water equivalent of the Karl Fischer reagent, g/ml.

W equals weight of $Na_2C_4H_4O_6\ 2H_2O$

V equals volume of Karl Fischer reagent used, ml.

b. Water-in-methanol solution. The water-in-methanol solution should contain 0.0015-0.0020 g of water per ml of solution. A good grade of commercial absolute methanol contains about 0.0010 g water per ml of methanol. Water content can be adjusted by adding 1.0 g of water to 1000 ml of the water-in-methanol solution to produce a change of 0.0010 g per ml. Determine the relative strength of the water-in-methanol solution in terms of Karl Fischer reagent as follows:

1. Put about 50 ml of the anhydrous methanol used in 4.8.5.2a into the titration beaker of the Aquameter. Add a slight excess of Karl Fischer reagent (4.8.5.2a), then back-titrate with water-in-methanol solution (4.8.5.2b). Then run in an additional 5 to 8 ml of Karl Fischer reagent, read to the nearest 0.01 ml, and again back-titrate with water-methanol solution (read to the nearest 0.01 ml). Repeat the addition and back-titrating steps twice more to provide triplicate determinations of the equivalency ratio. Calculate the ratio (B) of Karl Fischer reagent to that of the water-in-methanol solution. The range of the ratios calculated from the three titrations shall be not greater than 0.04. If the range exceeds 0.04, continue making titrations until three ratios are obtained whose range does not exceed 0.04. Then determine the average ratio from all the ratios which have been obtained.

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4.8.5.3 Procedure. Determine the moisture content of the sample by the Karl Fischer method using a direct-titration technique. Introduce to the titration beaker, through the opening in the diaphragm, approximately 10 g of sample weighed to the nearest 0.001 g and a mixture of 150 ml of anhydrous methanol with 50 ml of glacial acetic acid. Both materials are to be reagent grade. Close the opening, start the stirrer, and press the titrate button to titrate with Karl Fischer reagent (4.8.5.2a). When the indicator light glows, read the Karl Fischer buret to the nearest 0.01 ml. Where necessary, water-in-methanol solution (4.8.5.2b) may be used to back-titrate.

$$\text{Percent moisture equals } \frac{100A(V_{KF} \text{ minus } BV_{WM})}{W}$$

Where: A equals weight of water equivalent to 1.00 ml of Karl Fischer reagent, g/ml.

V_{KF} equals volume of Karl Fischer reagent titrant used, ml.

V_{WM} equals volume of water-in-methanol solution titrant used, ml.

B equals ratio of Karl Fischer reagent to that of water-in-methanol solution, ml/ml.

W equals weight of sample taken, g.

4.8.5.4 Acceptance criteria. For the lot represented to pass the moisture test, the value obtained for the percent moisture shall be no greater than the value specified in 3.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified by the contract or purchase order, preservation, packaging, packing, and marking requirements for the specified levels of protection shall be in accordance with MIL-P-14232.

6. NOTES

6.1 Intended use. N-phenylmorpholine described in this specification is intended for use as a stabilizer in ammonium-nitrate-based solid propellants.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of specification.

b. Whether a preproduction sample is required (see 3.1).

c. Type and size of shipping container (see 5.1.1).

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6.3 Definition.

6.3.1 Batch. A batch is defined as that quantity of material which has been subjected to one or more chemical or physical processes (or combinations thereof) intended to produce a desired product having substantially uniform characteristics.

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14159A, dated 23 December 1965.

Custodian:
ARMY-MI

Preparing Activity:
ARMY - MI
Project No. 6810-AA08

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